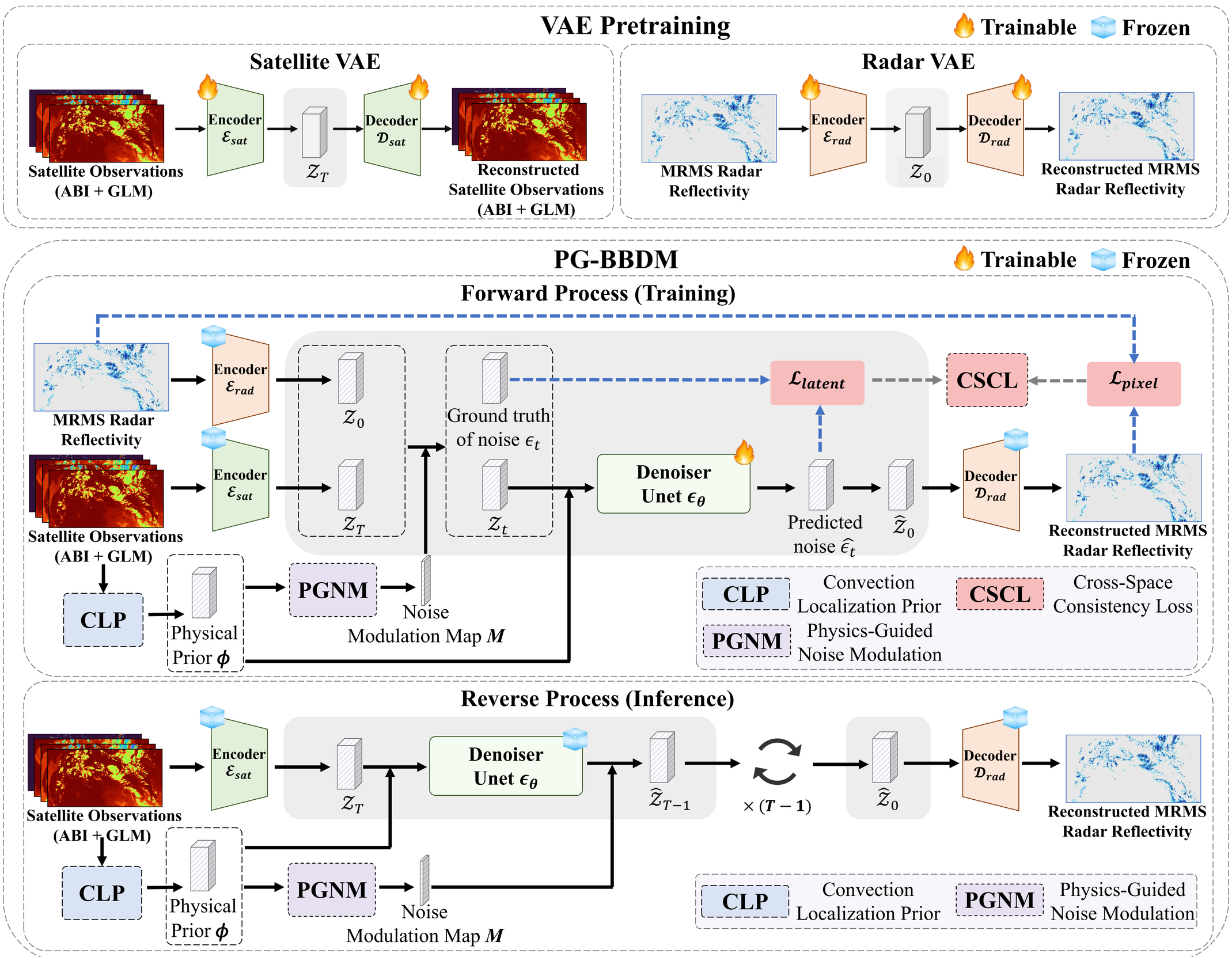
VAE Pretraining
Trainable
Frozen
Satellite VAE
Satellite Observations (ABI + GLM)
Encoder $\mathcal{E}_{sat}$
$\mathcal{Z}_T$
Decoder $\mathcal{D}_{sat}$
Reconstructed Satellite Observations (ABI + GLM)
Radar VAE
MRMS Radar Reflectivity
Encoder $\mathcal{E}_{rad}$
$\mathcal{Z}_0$
Decoder $\mathcal{D}_{rad}$
Reconstructed MRMS Radar Reflectivity
PG-BBDM
Trainable
Frozen
Forward Process (Training)
MRMS Radar Reflectivity
Encoder $\mathcal{E}_{rad}$
Satellite Observations (ABI + GLM)
Encoder $\mathcal{E}_{sat}$
$\mathcal{Z}_0$
$\mathcal{Z}_T$
Ground truth of noise ϵ_t
$\mathcal{Z}_t$
Denoiser Unet ϵ_θ
$\mathcal{L}_{latent}$
CSCL
$\mathcal{L}_{pixel}$
Predicted noise $\hat{\epsilon}_t$
$\hat{\mathcal{Z}}_0$
Decoder $\mathcal{D}_{rad}$
Reconstructed MRMS Radar Reflectivity
CLP
Physical Prior ϕ
PGNM
Noise Modulation Map M
CLP Convection Localization Prior
PGNM Physics-Guided Noise Modulation
CSCL Cross-Space Consistency Loss
Reverse Process (Inference)
Satellite Observations (ABI + GLM)
Encoder $\mathcal{E}_{sat}$
$\mathcal{Z}_T$
Denoiser Unet ϵ_θ
$\hat{\mathcal{Z}}_{T-1}$
$\times (T-1)$
$\hat{\mathcal{Z}}_0$
Decoder $\mathcal{D}_{rad}$
Reconstructed MRMS Radar Reflectivity
CLP
Physical Prior ϕ
PGNM
Noise Modulation Map M
CLP Convection Localization Prior
PGNM Physics-Guided Noise Modulation
z_0
z_1
z_2
Reverse Process
Forward Process
z_{T-2}
z_{T-1}
z_T